

Computer Vision assisted Deep Learning based Sitting Posture Suggestion System

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Abstract— This paper presents an innovative framework that employs camera-captured visual data to detect and suggest optimal sitting postures. The framework consists of two crucial components: a video capture object and an object detection system that incorporates Deep Learning to enhance efficiency and reliability. The camera initially captures the user's posture image, which is then subjected to video processing to extract video metadata. Subsequently, the object is created by extracting the image from the video, and the object detection algorithm is applied to provide posture recommendations. The algorithm continually monitors posture correctness and provides suggestions to improve it as necessary, ultimately benefiting the user's daily life and mitigating potential long-term problems. Additionally, the algorithm can be further developed to recognize posture patterns and suggest corrective exercises or techniques. Furthermore, the algorithm's efficiency can be enhanced by optimizing landmark detection for more effective outcomes. This cutting-edge framework offers immense potential for improving posture and overall health, and its development can significantly enhance the quality of life for individuals.

Keywords—Computer Vision, Sitting Posture Detection, Deep Learning, TensorFlow, MediaPipe, OpenCV

I. INTRODUCTION

With technology playing a crucial role in our daily lives, our health is often compromised by sedentary lifestyles, prolonged screen usage, and demands of modern work environments. Poor posture, a consequence of these factors, can result in various health issues, including musculoskeletal pain and decreased productivity. To solve this problem, innovative solutions combining advanced technologies with our understanding of human biomechanics are required. One such solution that has gained significant attention is posture detection using computer vision and deep learning. This emerging field utilizes computer vision to interpret and analyze visual data and deep learning to analyze images. With the fusion of these technologies, we can now teach computers to accurately perceive, recognize, and assess human postures. This technology has significant implications, from promoting healthier lifestyles and enhancing workplace ergonomics to revolutionizing healthcare and human-computer interaction. This paper proposes a framework for detecting and recognizing human poses in real-time using MediaPipe. MediaPipe offers

models for detecting landmarks on the face, hands, and body. The framework relies on OpenCV to process frames from a real-time device and export landmark coordinates to a CSV file. The paper focuses on detecting and recommending optimal sitting postures using visual data captured by a camera, which offers greater reliability and accuracy.

II. RELATED WORK

There have been numerous publications in the past detailing various methods for implementing sitting posture detection. Existing solutions for posture detection include orthopedic posture braces, skeleton-based posture recognition using depth sensors, wearable textile sensors, inertial and magnetic sensors, depth cameras, radio-frequency identification tags, 3D motion analysis, video surveillance, and sensors attached to office chairs [1]. These approaches rely on both colour and depth information, which may require additional hardware or sensors to capture the necessary data. [2] highlights the increased time people spend in front of computers leading to neck and back pains, which is supported by scientific research. It emphasizes the importance of proper monitoring of seated posture and taking breaks to prevent these pains. While the paper presents a lightweight posture guidance system with promising results, it lacks detailed information on the specific machine learning model, potential limitations of camera-based input, and considerations for implementation in different user contexts. [3] introduces a method and device for correcting a user's sitting posture by obtaining user information through a sensor on a mobile terminal and comparing it with the user's normal sitting posture information. A wristband with an internal sensor is used to monitor user posture and communicates with a mobile phone via Bluetooth. However, the paper fails to address any potential challenges or limitations in accurately calculating sitting posture information based on the distance and/or angle between the sensor and the user's key body parts. [4] presents a method for obtaining human body posture by accelerating OpenPose reasoning. The method involves obtaining an input video stream containing human body posture information, processing the single-frame image to obtain input data of an optimization model, reconstructing the model structure, reducing the precision of parameters in the OpenPose model, obtaining an optimization result, and finally obtaining human body posture data. This method cannot handle large-scale

datasets or real-time applications and is hence unscalable. [5] focuses on yoga pose detection using OpenPose and MediaPipe for image and video classification. It utilizes deep learning and neural network learning techniques for accurate pose recognition. The use of CNN and feature extraction methods enhances the accuracy of pose detection. [6] paper presents a novel approach for accurately estimating the pose of objects using deep learning. The algorithm consists of two primary stages: object detection and spatial reconstruction. Object detection is performed using a Convolutional Neural Network (CNN) called PoseNet, which has proven to be effective in detecting and localizing objects in an image. Spatial reconstruction is achieved by utilizing stereo correspondences and 3D reconstructing the spatial coordinates of multiple ORB features within the object's bounding box. [7] proposes a method for real-time human posture recognition using a deep neural network, which improves the comprehensive performance of intelligent video surveillance systems and enhances the real-time experience of users. It achieves this by extracting joint information and synthesizing the structure information of the human skeleton to form a feature vector, which is then used for classification recognition using a multi-classification support vector machine but the paper does not provide a detailed analysis of the performance metrics or evaluation results of the proposed method, making it difficult to assess its accuracy and effectiveness. Traditional methods for human posture recognition often rely on handcrafted features and may have limitations in accurately capturing complex motion and posture characteristics. Depth learning methods, on the other hand, can automatically learn relevant features from raw depth data, leading to improved accuracy and robustness in posture recognition tasks [8]. The lack of publicly available labelled human posture datasets with depth information can limit the training and evaluation of depth learning models. Depth learning methods may require significant computational resources, especially for training large-scale models on large datasets, which can be a limitation for resource-constrained systems. [9] presents a method for estimating human body posture using a deep three-dimensional convolutional neural network (CNN) and post-processing techniques. The method involves inputting and preprocessing a human body posture dataset video frame, expanding it into a maximized square, and predicting the posture using the trained CNN. The final estimation result is obtained through post-processing. The method utilizes a time and space-based approach, encoding time as a dimensionality in the 3D convolution operation. The goal is to capture human body joint positions from videos using a 2D monocular camera. The paper focuses on using a 2D monocular camera for capturing videos, but it does not address the potential limitations or challenges associated with using a single camera for estimating three-dimensional human body postures. [10] addresses the issue of poor body posture and its impact on health, such as back pain and fatigue, which can affect daily activities. It highlights the increasing problem of compromised upright posture due to prolonged sitting and the high cost of treating lower back pain. To address this, the paper proposes a wearable posture detection system in the form of a belt connected to a mobile application. The belt's sensors detect incorrect posture, triggering a real-time buzzer and sending a notification to the mobile app. The app provides users with a daily report on their posture and offers personalized yoga and exercise recommendations to rectify it. In this report, we present a

method to detect and suggest sitting posture using visual data from a camera, which is more reliable and accurate.

III. CONTRIBUTIONS

- The paper presents an innovative framework that uses camera-captured visual data to detect and suggest optimal sitting postures. The framework consists of a video capture object and an object detection system that incorporates Deep Learning to enhance efficiency and reliability.
- The paper focuses on detecting and recommending optimal sitting postures using visual data captured by a camera, which offers greater reliability and accuracy.
- The technology discussed in this paper has significant implications, from promoting healthier lifestyles and enhancing workplace ergonomics to revolutionizing healthcare and human-computer interaction.

IV. FLOWCHART

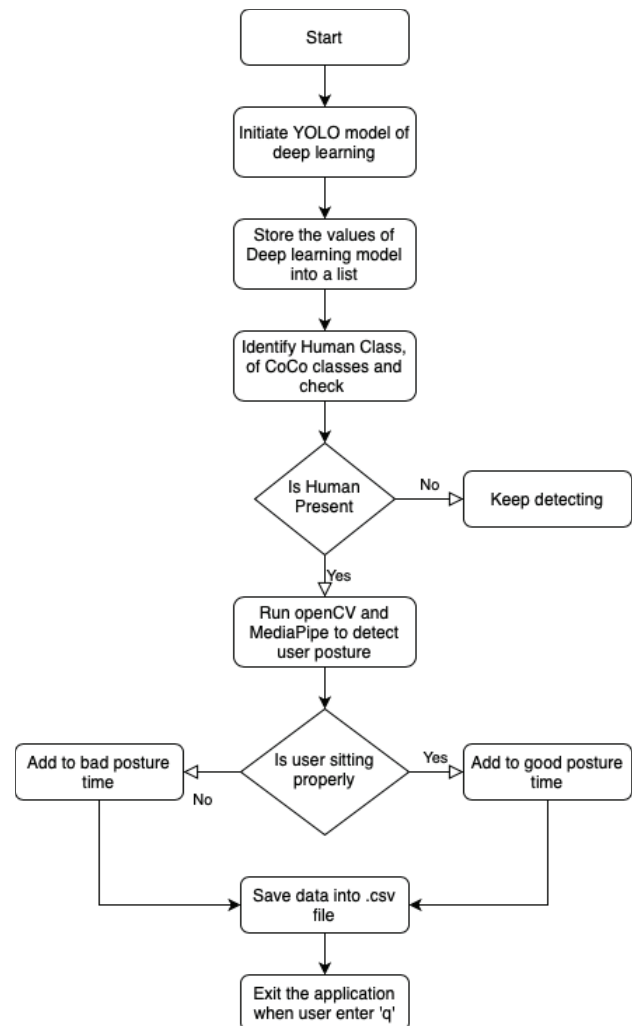


Fig. 1. Flow Chart of the proposed method

The flowchart presented in Fig. 1 offers a comprehensive overview of the system. We leverage the YOLO deep learning model for object detection, MediaPipe for pose estimation, and OpenCV for computer vision to achieve

accurate and valuable sitting posture detection. YOLO serves as the initial stage of the sitting posture detection system. YOLO is specially configured to identify two crucial objects: humans and chairs. The detection of both a human and a chair in the camera's field of view serves as a trigger for the posture assessment application. This ensures that the application activates only when the user is seated, with the presence of a chair. MediaPipe takes over once YOLO successfully identifies a human within the bounding box. It tracks and estimates the positions of these key body landmarks in real time. This step is crucial for accurately assessing the user's sitting posture. By continuously monitoring the movement and alignment of body landmarks, MediaPipe provides dynamic data for the posture assessment algorithm to work with. This real-time tracking ensures that users receive immediate feedback and guidance for maintaining a healthier and more ergonomic sitting posture. OpenCV, an open-source computer vision library, is at the heart of the project's image and video processing tasks. OpenCV performs a wide array of functions to enhance the video feed and facilitate effective posture assessment. It begins by capturing the video feed from the camera, enabling further analysis. OpenCV is responsible for preprocessing the frames, including resizing, cropping, and reducing noise, ensuring that the input data is of suitable quality for analysis. Furthermore, OpenCV excels in visualization, allowing the overlay of visual elements onto the video feed to provide real-time feedback to the user. These visualizations can take the form of graphical indicators, annotations, lines, or shapes. For example, lines may be drawn to represent the user's spine, emphasizing the angles of key body segments. OpenCV also handles the display of text-based feedback, presenting posture tips, warnings, or instructions tailored to the user's needs. Additionally, OpenCV offers real-time updates, enabling dynamic adjustments to the visual feedback as the user's posture changes.

V. BLOCK DIAGRAM

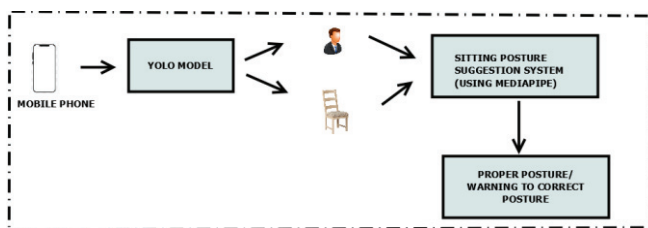


Fig. 2. Overall Block Diagram.

The block diagram presented in Fig. 2 provides a high-level view of the overall system. YOLO's initial detection of humans and chairs serves as the project's entry point, activating the application when both objects are present. This ensures that posture assessment occurs only when the user is seated with a chair. MediaPipe takes over after activation, estimating the positions of key body landmarks, while OpenCV processes the video feed for visualization and feedback. The three components work collaboratively to assess the user's sitting posture accurately and provide relevant real-time guidance. This integration ensures that the posture assessment is not only precise but also contextually relevant, making it a valuable tool for improving ergonomics and user well-being.

VI. PROPOSED METHOD

Import Libraries: Import the following libraries:

CV2: OpenCV, which stands for "Open Source Computer Vision Library," is a well-known open-source image processing and computer vision library that offers tools and functions for a variety of computer vision tasks. Computer vision is a branch of artificial intelligence that focuses on giving computers the ability to decipher and comprehend visual data from the outside world, such as pictures and videos.

Math: The math library is a built-in module that provides a wide range of mathematical functions and constants.

MediaPipe: MediaPipe is an open-source machine learning framework developed by Google that is primarily focused on enabling real-time, cross-platform, and efficient processing of media data, such as images and video streams. It offers a range of pre-trained machine learning models and instruments to carry out various computer vision and media analysis tasks. Because MediaPipe is so user-friendly, programmers may easily include machine learning capabilities in their applications.

YOLO: The popular deep learning method and object recognition technique known as YOLO, which stands for "You Only Look Once," is utilised in computer vision tasks. It is made to quickly and precisely identify and find items in pictures and videos. YOLO has made significant contributions to the field of object detection and has been widely adopted in both research and practical applications due to its speed and accuracy. It is commonly used in scenarios where real-time or near-real-time object detection is required.

NumPy: NumPy, short for "Numerical Python," is a core Python module for numerical and mathematical operations. It offers support in working with large, multidimensional arrays and matrices, as well as a range of mathematical operations to effectively interact with these arrays. Python's NumPy module is essential for scientific computing, data analysis, and machine learning.

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Function to calculate Distance: The findDistance function assists us in determining the offset distance between two places. It could be hip points, eyeballs, or the shoulder. These positions were chosen because they are always more or less symmetric about the human body's centre axis. We will include the camera alignment feature in the script as a result of this. We use the distance formula to calculate the distance between the body parts.

Function to calculate Body Posture Inclination: As shown in Fig. 3, the posture's main deterministic factor is the angle. We employ the angle that the torso line and neckline make with respect to the y-axis. The shoulder and the eye are connected via the neckline. Here, the shoulder is assumed to be the key axis. Similar to how the hip is a critical point, the torso line connects the hip and the shoulder.

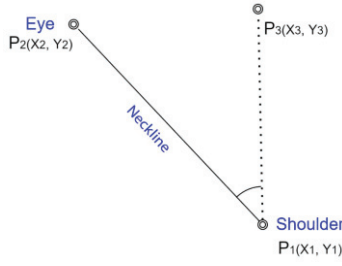


Fig. 3. Neckline inclination measurement

P1: shoulder

P2: eye

P3: Any vertical axis point that passes through P1

To determine the inner angle of three points, we use the vector method [11] as shown in Fig 4. two vectors $\vec{P_{12}}$ and $\vec{P_{13}}$'s angle is determined by:

$$\theta = \arccos\left(\frac{\vec{P_{12}} \cdot \vec{P_{13}}}{|\vec{P_{12}}| \cdot |\vec{P_{13}}|}\right)$$

Solving for θ we get,

$$\theta = \arccos\left(\frac{y_1^2 - y_1 \cdot y_2}{y_1 \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}}\right)$$

Vector angle Formula [11]

Function to send warning: This function is used to send a warning or alert when the wrong body posture is detected.

Apply Deep Learning Model: Import and save YOLO Deep Learning model as model.

Create Video Capture and Video Writer Object: Video capture and video writer objects are created. The video metadata is obtained to create the video capture object.

Processing the frames: The frames are captured through the webcam and their height and width are noted. Since the frames captured through the webcam are in BGR colour format, they are converted to RGB colour format and are then processed using the pose() method of the CV2 library. The frames are converted back to BGR colour format after processing.

Obtain Landmark Coordinates: The landmarks and normalised x and y coordinates are provided in the pose_landmarks attribute of the solution output object. The outputs must therefore be multiplied by the width and height of the image, respectively, in order to obtain the actual values. The PoseLandmark class's characteristics include the landmarks' LEFT_SHOULDER, RIGHT_SHOULDER, and so on.

Appending Angle Data: Data from angleData.csv file is reviewed and the live data captured is appended to the csv file.

Calculate Inclination and Draw landmarks: The function findAngle is used to determine the inclination angles. Landmarks and their connections are drawn.

Posture correction conditionals: Results are shown based on posture, whether it is good or bad. The frame counters for good and bad postures are increased with each detection, correspondingly. By dividing the number of frames by the frame rate (fps), one may determine the

duration of a certain position. The neck and torso inclination angle data is added to the csv file.

Output settings: The output is being configured to show results. Blue boxes highlight any humans or objects that are detected, and a message stating "Human Detected" appears when a person is detected. Following this, the posture detection algorithm is executed and provides the person with recommended posture suggestions.

VII. RESULTS AND OBSERVATIONS

Upon program execution, the webcam feed is captured and processed in real time. Initially, the program attempts to detect the presence of a human subject and displays a message when one is detected. The posture detection and suggestion algorithm is then activated.

Case 1: Good or Proper Posture:

If the user maintains good or proper posture, the algorithm highlights it in green and tracks its duration. This is illustrated in Fig. 5.



Good Posture Detected.

Case 2: Bad or Improper Posture:

If the user assumes an improper or bad posture, the algorithm identifies the specific area requiring correction and highlights it in red. Once corrected, the area turns green. If the user persists in an improper posture for over 3 minutes

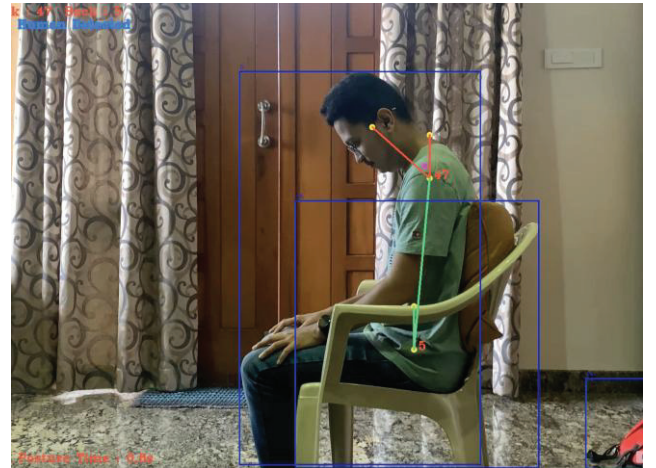


Fig. 4. Half Bad Posture Detected-1

despite suggestions, a warning is displayed. Additionally, the algorithm monitors the duration of correct posture and offers suggestions to improve it as needed.

As seen in Fig. 6, the subject's torso inclination is proper but his neck inclination is improper, hence only that part is highlighted in red and the rest in green.



Fig. 5. Half Bad Posture Detected-2.

In Fig. 7, it is seen that the subject's neck inclination is proper but his torso inclination is improper, hence only that part is highlighted in red and the rest in green, and when both the neck and torso inclinations of the subject are improper then the whole posture is highlighted in red as seen in Fig. 8, and if the user takes the algorithm's suggestions and corrects his posture, then his whole posture is highlighted in green as seen in Fig. 5.

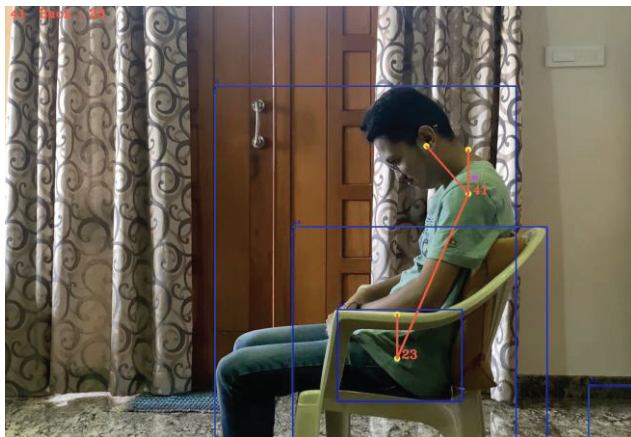


Fig. 6. Full Bad Posture Detected.

In Fig. 9, a graph is presented using the Spine dataset values that were obtained from Kaggle. This graph helps to identify the correct posture and the normal angle values. By referring to this graph, we can determine an optimal angle that serves as a benchmark to validate the user's position. The graph also indicates the maximum values for neck inclination and torso inclination that are considered normal. We then collect user data in multiple sessions and compare it with the standard dataset and graph shown in Fig. 10. The aim of collecting this data is to ensure that the user maintains a consistent posture, which can prevent back or neck problems in the long run.

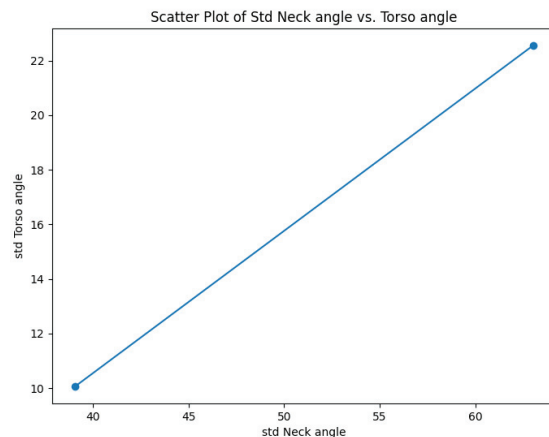


Fig. 7. Spine Dataset Plot

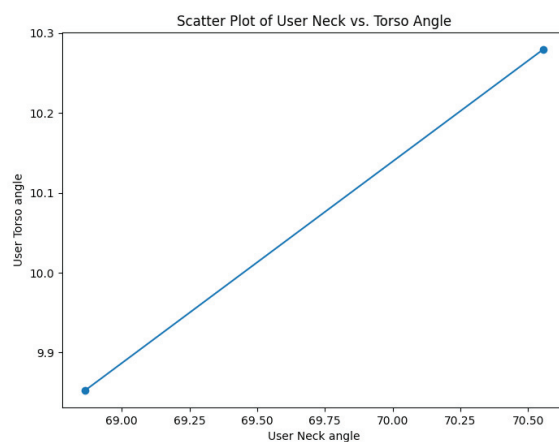


Fig. 8. Collected Angle Dataset Plot

VIII. CONCLUSION

In this paper, we delve into the possibilities of utilizing cutting-edge technologies such as Computer Vision, Machine Learning, and Deep Learning to effectively detect incorrect sitting posture. Our ultimate goal is to develop an application that can quickly identify improper posture and offer users corrective measures. The algorithm we discuss employs MediaPipe, a tool from Google, to identify key points such as the neck, shoulder, hip, and ears, and determine the angles of the neck and torso. Additionally, we have integrated an object detection system that uses Deep Learning to improve efficiency. After analyzing data collected from multiple sessions, the technology generates a report on the user's posture. By tackling the issue of incorrect sitting posture, our technology has the potential to alleviate long-term problems. The collected data is later compared to standard data in order to determine whether the subject may experience back problems in the future. The future direction of this system involves developing an application that can monitor posture, track the subject's posture duration, and generate a detailed report. This paper highlights the progress made in posture detection technology through the convergence of computer vision and machine learning, resulting in more precise and automated measurements compared to manual techniques. Future research could explore the integration of wearable devices or sensors with computer vision and deep learning methods to enable real-

time posture monitoring and feedback. Additionally, there is potential for developing more sophisticated algorithms and models to enhance the accuracy and reliability of posture detection systems, particularly in complex and dynamic settings. Investigating the feasibility of using posture detection systems in healthcare settings, such as preventing or monitoring musculoskeletal disorders or supporting rehabilitation programs, could be a promising area for future exploration. Conducting user studies and evaluations to evaluate the effectiveness and user acceptance of posture detection systems in various settings, including workplaces, educational institutions, and home environments, can provide valuable insights for further development and implementation.

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